

Loop Invariant and While Loop Rule

- What should be the $wp(W, q)$ for $W \equiv \text{while } B \text{ do } S \text{ od}$? Here let's assume that W is error-free.
 - Denote w_i as the weakest precondition where loop W runs exactly i iterations.
 - If we never enter the loop body, then $wp(W, q) = w_0 = \neg B \wedge q$.
 - If W runs exactly one iteration, then $wp(W, q) = w_1 = B \wedge wp(S, w_0)$
 - If W runs exactly two iterations, then $wp(W, q) = w_2 = B \wedge wp(S, w_1)$
 - ...
 - If W runs exactly $k > 0$ iterations, then $wp(W, q) = w_k = B \wedge wp(S, w_{k-1})$
 - In general, we cannot predict how many iterations are needed for a loop, thus $wp(W, q) \equiv w_0 \vee w_1 \vee w_2 \dots \vee w_k$ where k is the number of iterations W being executed, and k can be arbitrarily large. In other words, we cannot express $wp(W, q)$ with finite text.
 - The same problem also happens when we calculate $sp(p, W)$.
- Since we cannot calculate $wp(W, q)$ exactly, all we can do is to approximate it. In other words, we want to find a $p \Rightarrow wp(W, q)$, and p is not much stronger than $wp(W, q)$.
 - Here is an idea: if we can find a $p \Rightarrow w_i$ for each i , then $p \Rightarrow w_0 \vee w_1 \vee w_2 \dots \vee w_k$.
- A **loop invariant** for $W \equiv \text{while } B \text{ do } S \text{ od}$ is a predicate p such that $\models \{p \wedge B\} S \{p\}$. It follows that $\models \{p\} W \{p \wedge \neg B\}$.
 - In other words, a loop invariant is a predicate that is True, before and after each iteration of the loop.
 - To indicate the loop invariant, we write an additional clause in the program: $W \equiv \{\text{inv } p\} \text{while } B \text{ do } S \text{ od}$.
- **While Loop Rule:**
 1. $\{p \wedge B\} S \{p\}$
 2. $\{p\} \text{while } B \text{ do } S \text{ od} \{p \wedge \neg B\}$ loop 1

Or:

 1. $\{p \wedge B\} S \{p\}$
 2. $\{\text{inv } p\} \text{while } B \text{ do } S \text{ od} \{p \wedge \neg B\}$ loop 1
 - The precondition p and postcondition $p \wedge \neg B$ are not necessarily the weakest precondition and the strongest postcondition, but they are the "best" precondition and postcondition we can find for a provable triple.
- 1. Which of the following predicate p can be a loop invariant for $W \equiv \{\text{inv } p\} \text{while } k < n \text{ do } k := k + 1; s := s + k \text{ od}$
 - a. $p \equiv T$ Yes.
 - b. $p \equiv F$ Yes.
 - c. $p \equiv 0 \leq k \leq n$ Yes, $p \wedge B \Leftrightarrow 0 \leq k < n$, so after $k := k + 1$, we still have p
- From the above example, we can see that not all loop invariants provide us useful information: it cannot be too weak or too strong. If we have a loop $W \equiv \text{while } B \text{ do } S \text{ od}$, and we need to show $\models \{p_0\} W \{q\}$, then we need a loop invariant p such that:



- 1) $p_0 \Rightarrow p$
- 2) $\models \{p \wedge B\} S \{p\}$ (which implies $\models \{p\} \textbf{while } B \textbf{ do } S \textbf{ od } \{p \wedge \neg B\}$)
- 3) $p \wedge \neg B \Rightarrow q$

In future classes, we will discuss more on how to find a good loop invariant.

2. Show that p is a loop invariant for $W \equiv \textbf{while } k < n \textbf{ do } k := k + 1; s := s + k \textbf{ od}$, where $p \equiv 0 \leq k \leq n \wedge s = \text{sum}(0, k)$, and $\text{sum}(0, k) \equiv \sum_{i=0}^k i$. Find a q then create a formal proof for provable triple $\{p\} W \{q\}$.
 - o To show p is loop invariant, we need to prove triple $\{p \wedge k < n\} k := k + 1; s := s + k \{p\}$ is provable. We can create the following proof.

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|--|-----------------------------|
| 1. $\{p[s + k / s]\} s := s + k \{p\}$ | backward assignment |
| 2. $\{p[s + k / s] [k + 1 / k]\} k := k + 1 \{p[s + k / s]\}$ | backward assignment |
| 3. $\{p[s + k / s] [k + 1 / k]\} k := k + 1; s := s + k \{p\}$ | sequence 2,1 |
| 4. $p \wedge k < n \Rightarrow p[s + k / s] [k + 1 / k]$ | predicate logic |
| $\# p \wedge k < n \Leftrightarrow 0 \leq k < n \wedge s = \text{sum}(0, k)$ | |
| $\# p[s + k / s] [k + 1 / k] \equiv (0 \leq k \leq n \wedge s + k = \text{sum}(0, k)) [k + 1 / k]$ | |
| $\equiv 0 \leq (k + 1) \leq n \wedge s + (k + 1) = \text{sum}(0, k + 1)$ | |
| 5. $\{p \wedge k < n\} k := k + 1; s := s + k \{p\}$ | strengthen precondition 4,3 |
| 6. $\{p\} W \{p \wedge k \geq n\}$ | loop 5 |

3. Prove the following triple under partial correctness.

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{ $n \geq 0$ }
 $k := 0; s := 0;$ 
{inv  $p \equiv 0 \leq k \leq n \wedge s = \text{sum}(0, k)$ }
while  $k < n$  do
   $s := s + k + 1; k := k + 1$ 
od
{ $s = \text{sum}(0, n)$ }

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- o Informally, to prove the above program we need to prove the following triples/predicates:

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 $\{n \geq 0\} k := 0; s := 0 \{p\}$ 
 $\{p\} \textbf{while } k < n \textbf{ do } s := s + k + 1; k := k + 1 \textbf{ od } \{p \wedge k \geq n\}$ 
 $p \wedge k \geq n \Rightarrow s = \text{sum}(0, n)$ 

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We can create the following proof.

- | | |
|--|--------------------------|
| 1. $\{n \geq 0\} k := 0 \{n \geq 0 \wedge k = 0\}$ | forward assignment |
| 2. $\{n \geq 0 \wedge k = 0\} s := 0 \{n \geq 0 \wedge k = 0 \wedge s = 0\}$ | forward assignment |
| 3. $\{n \geq 0\} k := 0; s := 0 \{n \geq 0 \wedge k = 0 \wedge s = 0\}$ | sequence 1,2 |
| 4. $n \geq 0 \wedge k = 0 \wedge s = 0 \rightarrow p$ | predicate logic |
| $\# \text{Where } p \equiv 0 \leq k \leq n \wedge s = \text{sum}(0, k)$ | |
| 5. $\{n \geq 0\} k := 0; s := 0 \{p\}$ | weaken postcondition 3,4 |
| 6. $\{p[k + 1 / k]\} k := k + 1 \{p\}$ | backward assignment |
| 7. $\{p[k + 1 / k] [s + k + 1 / s]\} s := s + k + 1 \{p[k + 1 / k]\}$ | backward assignment |
| 8. $\{p[k + 1 / k] [s + k + 1 / s]\} s := s + k + 1; k := k + 1 \{p\}$ | sequence 7,6 |
| 9. $p \wedge k < n \rightarrow p[k + 1 / k] [s + k + 1 / s]$ | predicate logic |

$$\begin{aligned} & \# p \wedge k < n \Leftrightarrow 0 \leq k < n \wedge s = \text{sum}(0, k) \\ & \# p[k+1/k][s+k+1/s] \equiv (0 \leq k+1 \leq n \wedge s = \text{sum}(0, k+1)) [s+k+1/s] \\ & \quad \equiv 0 \leq k+1 \leq n \wedge s+k+1 = \text{sum}(0, k+1) \end{aligned}$$

10. $\{p \wedge k < n\} s := s + k + 1; k := k + 1 \{p\}$ strengthen precondition 9, 8

11. $\{p\} W \{p \wedge k \geq n\}$ loop 10

Where $W \equiv \mathbf{while} \ k < n \ \mathbf{do} \ s := s + k + 1; k := k + 1 \ \mathbf{od}$

12. $\{n \geq 0\} k := 0; s := 0; W \{p \wedge k \geq n\}$ sequence 5, 11

13. $p \wedge k \geq n \rightarrow s = \text{sum}(0, n)$ predicate logic

$\# p \wedge k \geq n \Leftrightarrow 0 \leq k = n \wedge s = \text{sum}(0, k)$

14. $\{n \geq 0\} k := 0; s := 0; W \{s = \text{sum}(0, n)\}$ weaken postcondition 12, 13